REAPS

Recommendation Engine for AI Adoption Policy and Strategy

Technical Report
SE - 801 - Software Project Lab III

Supervised by:
Dr. Md. Mahbubul Alam Joarder
Professor
Institute of Information Technology
University of Dhaka

Submitted by:
Mehzabin Haque (1233)
Institute of Information Technology
University of Dhaka

Submitted to:
SPL 3 Program Committee

Submission Date: December 1, 2024



Institute of Information Technology
University of Dhaka

Letter of Transmittal

1st December, 2024
BSSE 4th Year Exam Committee
Institute of Information Technology
University of Dhaka

Subject: Technical Report Submission - **REAPS** (Recommendation Engine for AI Adoption Policy and Strategy).

Sir,
I am submitting the technical report for our Software Project Lab 3, "**REAPS:** Recommendation Engine for AI Adoption Policy and Strategy".
This report comprehensively covers the project's technical details, including design, implementation, and testing phases. While I've tried to do my best, I welcome your feedback for improvement.

Thank you for your consideration.

Sincerely,
Mehzabin Haque
BSSE-1233
Institute of Information Technology
University of Dhaka

01.12.2024
——----------
Supervisor's Signature

Acknowledgement

I am grateful to the Institute of Information Technology for giving me the opportunity to work on the project "REAPS (Recommendation Engine for AI Adoption Policy and Strategy)". I would like to convey my tremendous gratitude to my supervisor, Dr. Md. Mahbubul Alam Joarder, Professor, Institute of Information Technology, University of Dhaka, for providing me with guidance and support throughout the project. His expertise and insights were instrumental in shaping the direction of the project. Lastly, I would like to thank my classmates. They have always been helpful and provided valuable insights from time to time.

Abstract

In this rapidly growing sphere of AI, governance and strategic policy formulation obtain considerable importance, ensuring their responsible and impactful deployment. However, it is often difficult for policymakers and analysts to conduct any comprehensive assessment of existing AI policies, pinpointing which are regularized, up-to-date, and of global quality. Several challenges have to be dealt with in making it happen. The REAPS recommendation engine is a highly advanced tool based on NLP techniques with LLMs. The program allows uploading an AI policy document and analyzes it automatically for gaps by comparing it against OECD and World Bank international standards. Informed decision-making and strategic planning are further facilitated by REAPS through AI readiness scorecards and actionable, tailored recommendations in an automated format. These may offer comparative insights into policies of peer nations, with a potential availing of best practices in the art of putting in place better regulatory frameworks. It allows for a friendly web-interface interaction during the uploading of documents, customizing the analysis, viewing the detailed results, and feedback for system improvement. REAPS enables discreet, evidence-based insights into smoother policy development by placing government officials, policy analysts, and researchers in a position to develop strong and effective governance frameworks for AI.

1. Introduction....1
1.1 Purpose....1
1.2 Motivation....1
1.3 Scope....2
1.4 Timeline....2
1.5 Purpose of Document....3
2. Project Description....4
2.1.1 Normal Requirements....4
2.1.2 Expected Requirements....5
2.1.3 Exciting Requirements....6
2.2 Usage Scenario for REAPS....6
Example Scenario Walkthrough....8
3. Scenario Based Modeling....9
3.1 Use Case Diagram....9
3.1.1 REAPS....9
3.1.2 Modules of REAPS....11
3.1.2.1 Sign-Up and Authentication....12
3.1.2.2 Run Policy Analysis....12
3.1.2.3 Policy Focus Customization....12
3.1.2.4 Subscribe to Updates....13
3.1.2.5 Compare Policies Across Countries....13
3.1.2.6 Access Readiness Scoreboard....13
4. Data Based Modeling....14
5. Class-Based Modeling....19
5.1 Analysis Classes....19
5.2 Class cards....20
6. Architectural Design....28
6.1 Architectural Context Diagram....28
6.2 Archetypes....30
6.3 Top Level Components....32
6.4 Deployment Diagram....32
7. Preliminary Test Plan....34
7.1 High Level Description of Testing Goals....34
7.2 Test Cases....34

Test Case 1: User Registration and Authentication..... 34
Test Case 2.1: Document Upload and Validation.....35
Test Case 2.2: Invalid Document Upload Handling.....35
Test Case 3: Country-Specific Policy Analysis.....35
Test Case 4: Receiving Tailored Recommendations.....36
Test Case 5: AI Readiness Scorecard.....36
Test Case 6: Comparative Insights with Similar Countries.....36
Test Case 7: User Customization of Preferences.....37
Test Case 8: Performance Under High Load..... 37
Test Case 9: Regular Policy Updates and Monitoring..... 37
Test Case 10.1: Validation with High-Income Countries for Similar Outputs..38
Test Case 10.2: Validation with Low-Income Countries for Similar Outputs.. 38
Test Case 10.3: Cross-Validation Between High-Income and Low-Income Countries.....39
Test Case 11: Divergence Validation for Different Income Levels.....39
Test Case 12: Validation Against Known Benchmarks..... 39
Test Case 13: Testing Recommendations' Customization.....40
8. Conclusion.....41
References..... 42

List Of Figures

Fig-1: Timeline for REAPS 3
Fig-2: Use Case Level 0 (REAPS). 10
Fig-3: UseCase Level 1 (REAPS) 13
Fig-4: ER Diagram of REAPS 15
Fig-5: CRC Diagram of REAPS 27
Fig-6: Architectural Context Diagram of REAPS 28
Fig-7: Archetype of REAPS 30
Fig-8: Top Level Components 32
Fig-9: Deployment diagram of REAPS 33

List of Tables

Table-1: Schemas of REAPS 16
Table-2: CRC Card (User) 20
Table-3: CRC Card (PolicyDocument) 21
Table-4: CRC Card (PolicyAnalysis) 21
Table-5: CRC Card (Recommendation) 22
Table-6: CRC Card (ReadinessScorecard) 23
Table-7: CRC Card (Notification) 23
Table-8: CRC Card (Country) 24
Table-9: CRC Card (PolicyGap) 24
Table-10: CRC Card (ComparativeInsight) 25
Table-11: CRC Card (GlobalBenchmark) 25
Table-12: CRC Card (LLMService) 26

1. Introduction

REAPS is an AI-based recommendation engine developed to advise governments on adaptive, context-specific, and progressive AI policies. With a keen eye on great changes happening in the world-variations of various socio-economic and cultural factors. REAPS aims to provide customized modes of policy recommendations on AI policy. Some nations, like Bangladesh and the USA, have completely different priorities, necessitating policies pertaining to their unique requirements. By applying machine learning to vast quantities of policy data, REAPS dynamic facilitates strategizing according to the unique backgrounds of various zones.

1.1 Purpose

The main reason why REAPS exists is to help governments to make policies around AI that are flexible, progressive, and tailored to their unique socio-economic, cultural and technological contexts. Through the examination of policy documents, REAPS generates bespoke recommendations and identifies policy gaps and thus enables governments to harmonize their own AI policies with international norms while responding to local issues and opportunities. It also intends making policy making responsive to the latest global trends, benchmarks and best practices in AI government, for adopting responsible and efficient AI at a national level.

1.2 Motivation

The motivation behind REAPS originates from the increasing relevance that artificial intelligence is assuming for transforming and driving the economies of societies. Indeed, there is a dazzling promise for AI to change the face of industries. However, for that potential to be realized, the governments of the countries concerned must possess strategic, forward-looking, and strong AI policies. Unfortunately, many nations, especially those of the developing world, have the problem of limited resources, knowledge, and access to global trends, which makes it tough for them to put sound national policies in place on AI.

REAPS hopes to democratize access to world-class AI policy insights by providing a tool through which governments can make data-backed decisions. It also offers an

AI-supported recommendation engine that will synergize policymaking with the best practices so that, like Bangladesh, competing countries benefit from AI advancements. It is particularly important for countries by providing proactive solutions instead of piecemeal or reactive national AI strategies, with particular emphasis on using global trends.

1.3 Scope

REAPS is designed to:

1. Analyze national AI policies, identify gaps, and assess alignment with global standards.
2. Provide customized, context-specific AI policy recommendations.
3. Generate AI readiness scorecards and monitor policy progress over time.
4. Support stakeholders with data-driven insights for sustainable and ethical AI adoption.
5. Ensure scalability for multiple countries and adaptability to evolving AI governance trends.

1.4 Timeline

A 4-month timeline has been planned for this project.

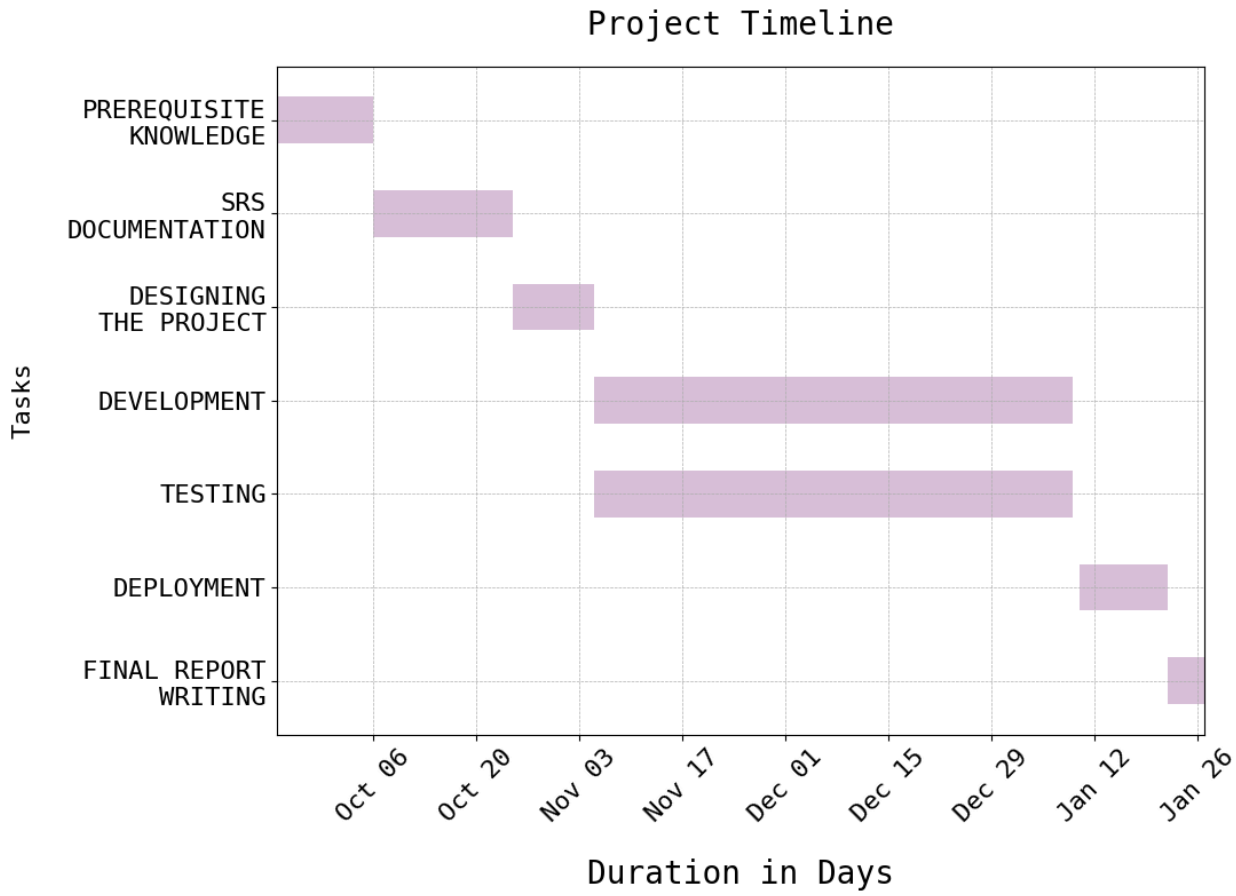


Fig-1: Timeline for REAPS

1.5 Purpose of Document

The purposes of this document are-

- Identify and analyze the requirements
- Design the architecture
- Design the test plan
- Describe methodology
- Reduce the development effort
- Improve understanding

2. Project Description

The development of REAPS commenced with a comprehensive review of research literature on AI policy frameworks, international governance standards, and socio-economic impacts of AI adoption. Key references included benchmark studies on global AI readiness indices and ethical AI guidelines, which offered critical insights into the domain. This foundation was further strengthened by proactive engagement with stakeholders, including government policymakers, research institutions, think tanks, and international organizations such as the OECD and World Bank. These interactions provided diverse perspectives, addressing challenges such as policy gaps, ethical concerns, and governance needs in AI policy-making.

To enable sophisticated analysis and recommendations, REAPS integrates advanced Large Language Models (LLMs). These models perform natural language processing on uploaded policy documents, extracting key information and identifying areas for improvement. Additionally, data from international organizations provide authoritative benchmarks and socio-economic indicators, ensuring that REAPS' recommendations are aligned with global standards.

This collaborative and technology-driven approach shaped REAPS into a robust, adaptive solution tailored to the unique needs of diverse socio-economic contexts, empowering policymakers and researchers with actionable insights.

2.1.1 Normal Requirements

Normal requirements encompass the fundamental objectives and functionalities that are explicitly stated by customers. Meeting these ensures baseline satisfaction and the tool's core utility.

Functional Requirements:

- **Standard Policy Recommendations:** Provide general AI policy suggestions based on analyzed data. Ensure recommendations align with basic regulatory and ethical standards.

- **Country Specific Policy Analysis:** Generate AI policy recommendations specifically aligned with a country's socio-economic and cultural characteristics.
- **Policy Analysis from Documents:** Implement algorithms to accurately extract data from various AI policy documents, regardless of format. Conduct fundamental analysis to identify the structure, key components, and overarching themes within existing AI policies.

Non-Functional Requirements:

- **Usability:** Design a user-friendly interface that guides users through document upload, country selection, and viewing recommendations seamlessly.

2.1.2 Expected Requirements

Expected requirements are intrinsic to the product and, while not always explicitly stated by customers, their absence can lead to significant dissatisfaction.

Functional Requirements:

- **Validation Mechanism:** Compare recommendations with AI policies of socio-economically similar countries to ensure relevance and accuracy.
- **Regular Policy Updates and Monitoring:** Provide quarterly or annual insights on evolving AI policies, new benchmarks, and emerging practices.
- **AI Readiness Scorecard:** Generate a scorecard indicating a country's AI readiness based on current policies and implementation progress.
- **Comprehensive Data Integration:** Integrate data from multiple reliable sources, including government websites, legal repositories, and academic journals.

Non-Functional Requirements:

- **Reliability:** Maintain dependable performance, especially in report generation and regular updates.

2.1.3 Exciting Requirements

Exciting requirements exceed customer expectations, adding innovative features that enhance user satisfaction and differentiate the product.

Functional Requirements:

- **User Customization:** Allow users to customize the frequency of updates, select preferred countries for peer comparison, and focus on specific areas within AI policy (e.g., ethics, innovation, governance).
- **Interactive Dashboards:** Offer interactive dashboards that allow users to explore data and recommendations visually.

2.2 Usage Scenario for REAPS

REAPS: AI Policy Recommendation and Evaluation Tool will be a web-based platform designed to help users, including policymakers and researchers, evaluate and improve AI policies. It provides tailored recommendations and readiness assessments based on uploaded policy documents and socio-economic contexts. REAPS will offer actionable insights and real-time reports to help users align their policies with global benchmarks. The usage scenario for REAPS can be segmented as follows:

Access

Users will access REAPS through a secure web portal. New users (e.g., government policymakers or researchers) can create an account and verify their credentials through email or organization-based authentication. Returning users can log in to their personalized dashboard, which serves as a central hub for policy analysis, recommendations, and updates.

Document Upload and Preprocessing

Users upload AI policy documents in supported formats such as PDF or DOCX. These documents could include draft policies, national strategies, or reports. REAPS preprocesses these documents securely, extracting key information using

advanced natural language processing (NLP) techniques powered by Large Language Models (LLMs). These models enable REAPS to understand the nuanced language of policy documents, identify key themes, and detect policy gaps.

Once a document is uploaded and the country is specified, REAPS tailors its analysis and recommendations by leveraging data from international organizations such as the OECD and World Bank. This data provides authoritative benchmarks on global AI policies, socio-economic indicators, and best practices. REAPS compares the uploaded policy against these benchmarks and similar policies from other countries to generate insights.

Country-Specific Policy Analysis

Once a document is uploaded and the country is specified, REAPS tailors its analysis and recommendations to align with the socio-economic and cultural context of the selected country. It identifies - policy gaps in governance, ethics, innovation, and economic impact, and actionable recommendations to improve policies and align them with global AI standards.

Real-Time Notifications and Recommendations

REAPS provides real-time feedback once the analysis is complete. Users receive - immediate recommendations based on the document and country context, notifications on detected policy gaps or areas needing improvement, and comparative insights with policies from similar countries or global benchmarks.

AI Readiness Scorecard

REAPS generates a detailed AI readiness scorecard for the selected country, which includes - a numerical readiness score based on global benchmarks, highlights of strengths and weaknesses in the current policy and a summarized evaluation guiding potential improvements in governance and implementation.

Interactive Dashboard and Insights

The REAPS dashboard offers a user-friendly interface where users can:

- Review uploaded documents and associated analysis reports.
- Access tailored recommendations and readiness scorecards.
- Subscribe to updates and notifications on global AI benchmarks and trends.

- Customize focus areas for analysis, such as ethics, governance, or innovation.

Example Scenario Walkthrough

User: A policy analyst from the Ministry of Innovation in Bangladesh.

1. **Sign-In:** The analyst signs into REAPS using their government-issued credentials.
2. **Upload Document:** They upload the country's latest draft of AI policy. REAPS validates the document and confirms it as a suitable AI policy document.
3. **Country Selection:** The analyst specifies Bangladesh to tailor the analysis.
4. **Running Analysis:** They run the analysis, and REAPS utilizes advanced LLMs to perform NLP on the document, extracting key information and identifying policy gaps. REAPS also incorporates data from the OECD and World Bank to compare Bangladesh's policy against global benchmarks and similar countries.
5. **Viewing Recommendations:** REAPS provides a report with tailored recommendations, highlighting areas like governance, ethical concerns, and economic implications, and suggests improvements based on international best practices.
6. **Viewing Readiness Scorecard:** The analyst checks the AI Readiness Scorecard, which incorporates international benchmarks, and finds that governance and ethics need further development.
7. **Subscription to Updates:** The analyst subscribes to quarterly updates for the latest global AI policy trends, which include data from international organizations.

Result: Equipped with tailored insights generated by LLMs and updated benchmarks from international organizations, the analyst refines their AI policy draft, addressing the identified gaps and aligning it with both local and global standards.

3. Scenario Based Modeling

Scenario-based modeling is an approach used in software development and design to create models that depict how a system behaves or reacts in various real-world situations or scenarios. These scenarios are typically based on user interactions, events, or conditions and help developers understand and visualize how the software will function in different contexts. By simulating these scenarios, developers can better identify potential issues, make informed design decisions, and ensure that the software meets user requirements across different usage contexts.

3.1 Use Case Diagram

A use case diagram is a visual representation in software engineering that illustrates how a system interacts with its external entities or actors to achieve specific goals or tasks. It shows the relationships between use cases (representing system functionalities) and actors (representing external users or systems). Use case diagrams are typically created through requirements gathering, where system functionalities and user interactions are identified, and then diagrammed using specialized software modeling tools.

3.1.1 REAPS

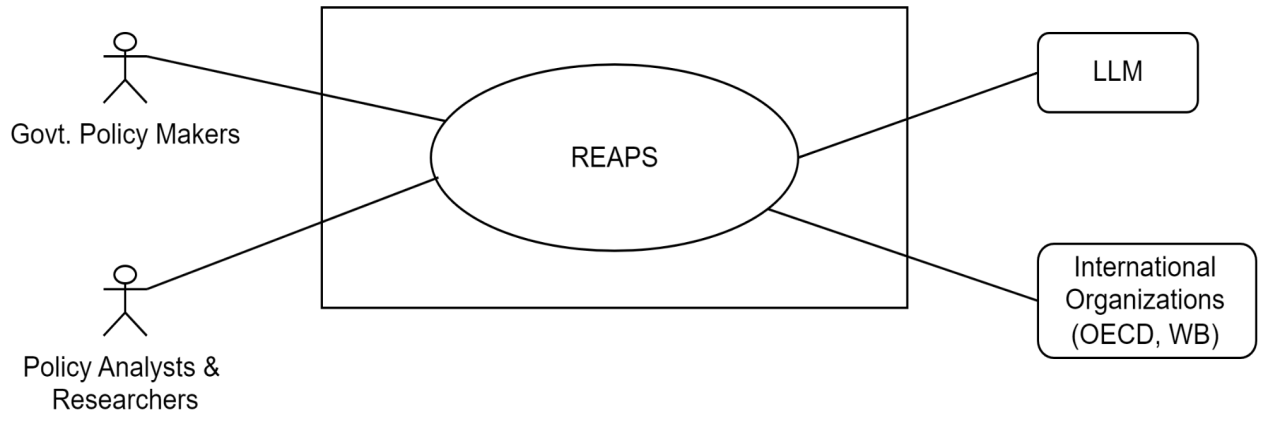


Fig-2: Use Case Level 0 (REAPS).

Primary actors: Govt. Policy Makers, Policy Analysts & Researchers
Secondary actors: International Organizations (OECD, WB), LLM
Goal in Context: The diagram shown above represents the Recommendation Engine for AI Adoption Policy and Strategy (REAPS)

3.1.2 Modules of REAPS

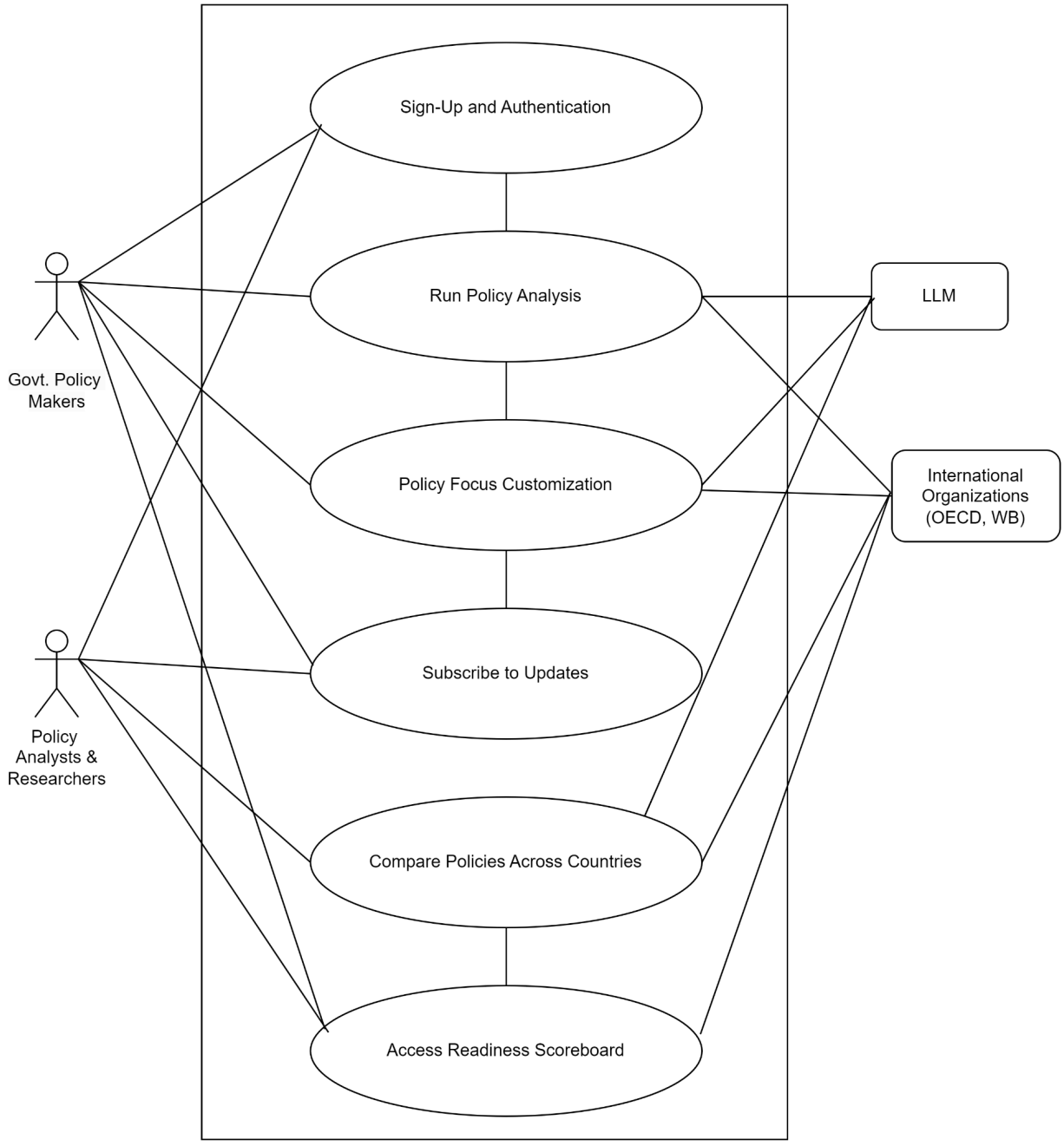


Fig-3: UseCase Level 1 (REAPS)

Primary actors: Govt. Policy Makers, Policy Analysts & Researchers
Secondary actors: International Organizations (OECD, WB), LLM

Goal in Context: The diagram shows the modules of the Recommendation Engine for AI Adoption Policy and Strategy (REAPS)
The use case can be elaborated as follows -

3.1.2.1 Sign-Up and Authentication

The "Sign-Up and Authentication" module serves as the gateway to REAPS' functionalities. It is responsible for verifying user credentials and granting access to authorized users. This module allows new users, such as policymakers and researchers, to create accounts with secure authentication and also differentiates access levels for various stakeholders, such as government officials and researchers.

3.1.2.2 Run Policy Analysis

The "Run Policy Analysis" module is central to REAPS, allowing users to initiate comprehensive analyses of uploaded AI policy documents in formats like PDF or DOCX. Utilizing advanced Natural Language Processing (NLP) techniques powered by LLMs, the module extracts key information and performs complex analyses such as sentiment analysis, topic modeling, and entity recognition. It benchmarks the policies against global AI governance standards and data from international organizations like the OECD and World Bank, identifying policy gaps in areas such as ethics, technological readiness, and regulatory frameworks. The module then generates tailored recommendations that align with the country's socio-economic and cultural context, enhancing policy effectiveness and adherence to international best practices.

3.1.2.3 Policy Focus Customization

The "Policy Focus Customization" module empowers users to tailor the analysis process according to their specific priorities. Users can select focus areas such as ethics, innovation, governance, or economic impact, allowing for detailed and targeted analyses. This customization adjusts the parameters of the LLMs to emphasize user-defined priorities, resulting in more relevant and actionable recommendations and insights. An intuitive interface facilitates easy selection and modification of focus areas, while LLMs assist users by providing explanations and suggestions, ensuring that the analysis outputs are closely aligned with their unique needs and interests.

3.1.2.4 Subscribe to Updates

The "Subscribe to Updates" module sends the users periodic updates about evolving AI benchmarks, trends, and policy standards. Allows users to select the frequency of updates based on their needs. Focuses updates on user-specified areas of interest, such as ethics, AI readiness, or regulatory compliance.

3.1.2.5 Compare Policies Across Countries

The "Compare Policies Across Countries" module enables users to conduct comparative analyses of AI policies from multiple countries. Users can select countries with similar socio-economic profiles or from different regions to facilitate meaningful comparisons. Leveraging data from international organizations, the module analyzes differences and similarities in governance, innovation, readiness, and alignment with global best practices. A validation system assesses socio-economic similarities to ensure the accuracy and reliability of recommendations. LLMs generate comparative insights, highlighting areas where a country's policy aligns or deviates from others, providing detailed explanations and contextual information to support informed policy enhancements.

3.1.2.6 Access Readiness Scoreboard

The "Access Readiness Scoreboard" module offers a measurable assessment of a country’s AI readiness. It assigns a readiness score based on the UNESCO’s report. Tracks changes in readiness scores over time, allowing users to measure the impact of implemented policies.

4. Data Based Modeling

Data based modeling is a visual representation of a database structure and how data is organized within it. It involves creating entity-relationship diagrams (ERDs) that depict tables (entities), attributes (columns), and the relationships between them. Database modeling is typically achieved by analyzing the data requirements of a system, identifying entities and their attributes, and then using modeling tools or software to create ERDs that serve as a blueprint for database design and implementation.

The entities identified for REAPS are:

- Users
- AI Policy Documents
- Policy Recommendations
- Policy Updates
- Scorecards
- Country

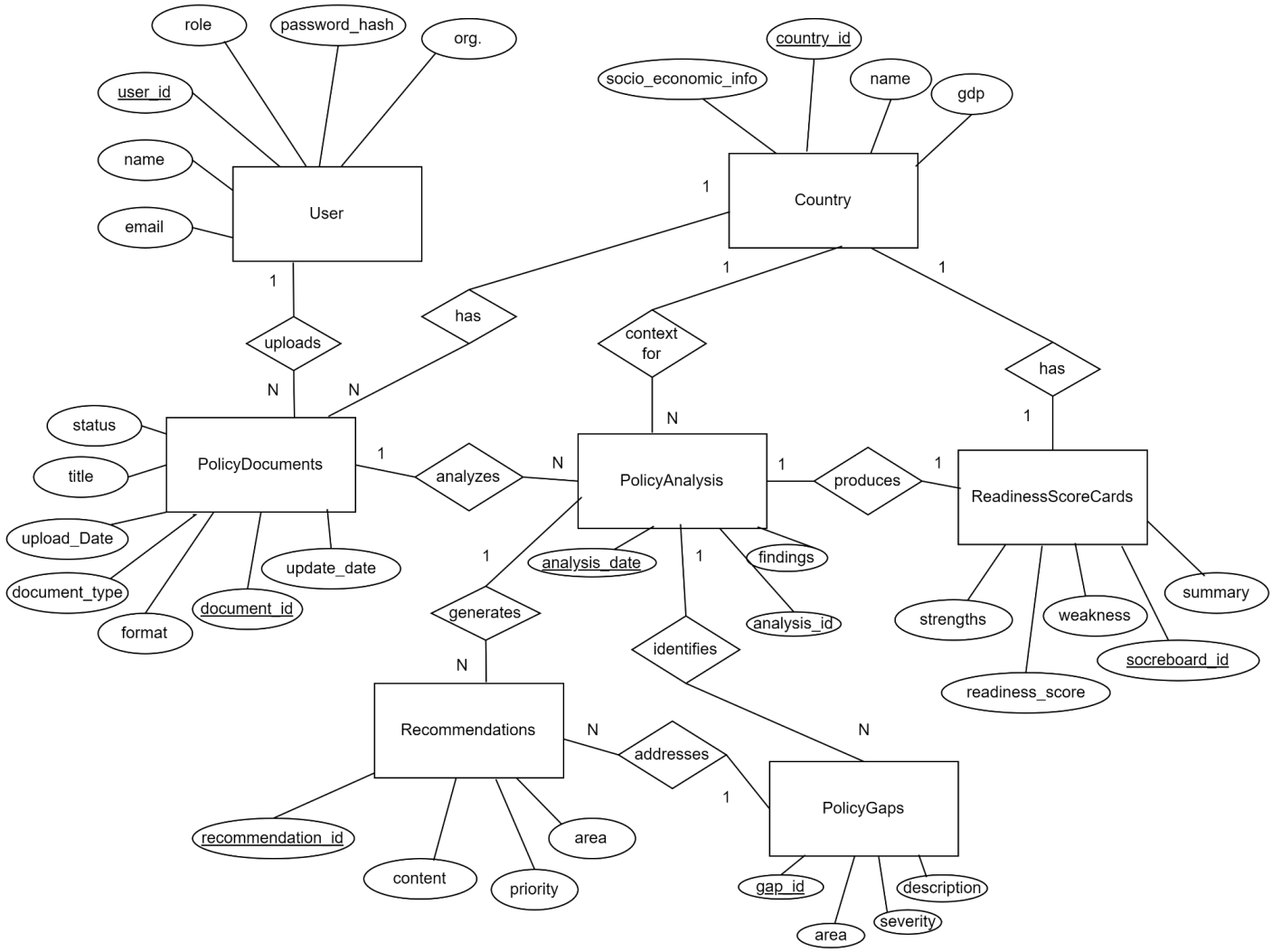


Fig-4: ER Diagram of REAPS

Table – 1: Schemas of REAPS

Table	Attribute Name	Data Type
User	user_id **(PK)**	INT
	name	VARCHAR(255)
	email	VARCHAR(255)
	password_hash	VARCHAR(255)
	organization	VARCHAR(255)
	role	ENUM('policymaker', 'researcher', 'admin')
Country	country_id **(PK)**	INT
	name	VARCHAR(255)
	gdp	DECIMAL(15,2)
	socio_economic_info	TEXT
PolicyDocuments	document_id **(PK)**	INT
	user_id **(FK)**	INT
	country_id **(FK)**	INT
	title	VARCHAR(255)
	content	TEXT
	upload_date	DATETIME
	status	ENUM('pending', 'analyzed')
	document_type	ENUM('draft_policy', 'strategy', 'report')

	format	ENUM('PDF', 'DOCX')
PolicyAnalysis	analysis_id **(PK)**	INT
	document_id **(FK)**	INT
	country_id **(FK)**	INT
	analysis_date	DATETIME
	findings	TEXT
Recommendations	recommendation_id **(PK)**	INT
	analysis_id **(FK)**	INT
	policy_gap_id **(FK)**	INT
	content	TEXT
	area	ENUM('governance', 'ethics', 'innovation', 'other')
	priority	ENUM('high', 'medium', 'low')
ReadinessScoreCards	scorecard_id **(PK)**	INT
	countrie_id**(FK)**	INT
	analysis_id(**FK**)	INT
	readiness_score	DECIMAL(5,2)
	strengths	TEXT
	weaknesses	TEXT
	summary	TEXT
PolicyGaps	gap_id **(PK)**	INT
	analysis_id **(FK)**	INT

	description	TEXT
	area	ENUM('governance', 'ethics', 'innovation', 'other')
	severity	ENUM('critical', 'moderate', 'minor')

5. Class-Based Modeling

Class-based modeling is a visual representation in software engineering that illustrates the structure of a software system using classes, their attributes, and relationships. It is commonly used in object-oriented design and modeling.

5.1 Analysis Classes

After identifying nouns from the scenario, I have filtered nouns belonging to the solution domain using General Classification (External entities, Things, Events, Roles, Organizational units, Places, and Structures). Nouns selected as a potential class were filtered using Selection Criteria (Retained information, Needed services, Multiple attributes, Common attributes, Common operations, and Essential requirements). After performing analysis on potential classes, I have found the following analysis classes

Based on REAPS' description, the key nouns include:

1. User
2. PolicyDocument
3. Analysis
4. Recommendation
5. ReadinessScorecard
6. Notification
7. Dashboard
8. Country
9. PolicyGap
10. ComparativeInsight
11. FocusArea
12. GlobalBenchmark
13. LLM

Using General Classification and Selection Criteria and after analysis, we have seen that similar responsibilities are being carried out by more than one class. So, classes with similar responsibilities are merged.

Derived Classes and Final Grouping -

1. User
2. PolicyDocument
3. PolicyAnalysis
4. Recommendation
5. ReadinessScorecard
6. Notification
7. Country
8. PolicyGap
9. ComparativeInsight
10. GlobalBenchmark
11. LLMService

5.2 Class cards

Table-2: CRC Card (User)

User	
Attributes	**Methods**
-userId -email -name -organization -userType -passwordHash -preferences -authenticated	-createAccount() -login() -logout() +uploadDocument(document) +selectCountry(countryID) -customizePreferences(preferences) +subscribeToUpdates(frequency) -receiveNotifications()
Responsibilities	**Collaborators**
• Manage user authentication and sessions. • Store and handle user-specific data and preferences. • Interface with policy documents and receive analysis results. • Subscribe to updates and notifications.	• PolicyDocument • Notification • Recommendation • Country • PolicyAnalysis

Table-3: CRC Card (PolicyDocument)

PolicyDocument	
Attributes	**Methods**
-documentID -userID -countryID -content -uploadDate -status -documentType -format	+upload() -preprocess() -validate() -associateWithCountry(countryID) +getContent()
Responsibilities	**Collaborators**
• Store and manage policy documents uploaded by users. • Preprocess and validate documents for analysis readiness. • Provide document content for policy analysis.	• User • PolicyAnalysis • Country • LLM

Table-4: CRC Card (PolicyAnalysis)

PolicyAnalysis	
Attributes	**Methods**
-analysisID -policyDocumentID -countryID -analysisDate -findings -recommendations -readinessScorecard -comparativeInsights	-runAnalysis() -generateFindings() -generateRecommendations() -generateReadinessScorecard() -generateComparativeInsights()
Responsibilities	**Collaborators**

• Perform comprehensive analysis of policy documents. • Generate findings, recommendations, and scorecards. • Provide real-time feedback and insights to users. • Ensure analysis is tailored to the country's context.	• PolicyDocument • Recommendation • ReadinessScorecard • PolicyGap • ComparativeInsight • GlobalBenchmark • Country • LLM

Table-5: CRC Card (Recommendation)

Recommendation	
Attributes	**Methods**
-recommendationID -analysisID -content -area -priority	-generate(content, area, priority) +getContent() +setPriority(level)
Responsibilities	**Collaborators**
• Provide actionable and tailored recommendations to users. • Categorize recommendations by policy area and priority. • Assist users in improving their AI policies.	• PolicyAnalysis • PolicyGap • User • Country • LLM

Table-6: CRC Card (ReadinessScorecard)

ReadinessScore	
Attributes	**Methods**
-scorecardID -analysisID -readinessScore -strengths -strengths -summary	-calculateScore() -generateSummary() +getStrengths() +getWeaknesses()
Responsibilities	**Collaborators**
• Assess and represent the AI readiness level of a country. • Highlight policy strengths and weaknesses. • Provide a summarized evaluation to guide improvements.	• PolicyAnalysis • GlobalBenchmark • Country

Table-7: CRC Card (Notification)

Notification	
Attributes	**Methods**
-notificationID -userID -content -type -date -readStatus	-sendNotification() -markAsRead() +getContent()
Responsibilities	**Collaborators**
• Deliver real-time updates and alerts to users. • Notify users about analysis	• User • PolicyAnalysis • Recommendation

results, updates, and recommendations. • Manage the notification status and history.	

Table-8: CRC Card (Country)

Country	
Attributes	**Methods**
-countryID -name -socioEconomicData -existingPolicies	+getSocioEconomicData() +getExistingPolicies() -compareWithGlobalBenchmarks
Responsibilities	**Collaborators**
• Provide country-specific context for policy analysis. • Store and manage country data relevant to AI policy. • Facilitate comparisons with global benchmarks.	• PolicyDocument • PolicyAnalysis • GlobalBenchmark • ComparativeInsight

Table-9: CRC Card (PolicyGap)

PolicyGap	
Attributes	**Methods**
-gapID -analysisID -description -area -severity	-identify(description, area, severity) +getDescription() +getSeverity()
Responsibilities	**Collaborators**
• Identify and describe deficiencies in the policy document.	• PolicyAnalysis • Recommendation

• Categorize gaps by area and severity. • Provide information to generate recommendations.	• LLM

Table-10: CRC Card (ComparativeInsight)

ComparativeInsight	
Attributes	**Methods**
-insightID -analysisID -similarCountries -similarities -differences -validationScore	-generateComparativeAnalysis() +getSimilarities() +getDifferences() -calculateValidationScore()
Responsibilities	**Collaborators**
• Compare the country's policies with those of similar nations. • Validate the relevance and accuracy of recommendations. • Provide insights for policy enhancement based on comparisons.	• PolicyAnalysis • Country • GlobalBenchmark • LLM

Table-11: CRC Card (GlobalBenchmark)

GlobalBenchmark	
Attributes	**Methods**
-benchmarkID -area -standards -lastUpdated -updates	+getStandards(area) -updateStandards -compareWithPolicy(policyContent)

Responsibilities	**Collaborators**
• Provide up-to-date global standards and benchmarks. • Facilitate comparison of policies with international best practices. • Keep track of evolving AI policy trends and standards.	• PolicyAnalysis • Country • ReadinessScorecard • ComparativeInsight • LLM

Table-12: CRC Card (LLMService)

LLMService	
Attributes	**Methods**
-modelName -version -apiEndpoint -parameters -contextData	-processDocument(documentContent) -generateAnalysis(documentContent, countryData) -generateRecommendations(analysisResults, focusAreas) -summarizeData(data) -assistUser(query) -performNLPTasks(taskType, content)
Responsibilities	**Collaborators**
• Perform natural language processing on policy documents. • Generate policy analyses tailored to specific countries. • Produce actionable recommendations based on analysis results. • Summarize complex data into digestible insights. • Provide intelligent assistance to users during interactions. • Enhance other components by providing NLP capabilities	• PolicyAnalysis • PolicyDocument • Recommendation • ComparativeInsight • PolicyGap • GlobalBenchmark

The CRC diagram of the classes is given below:

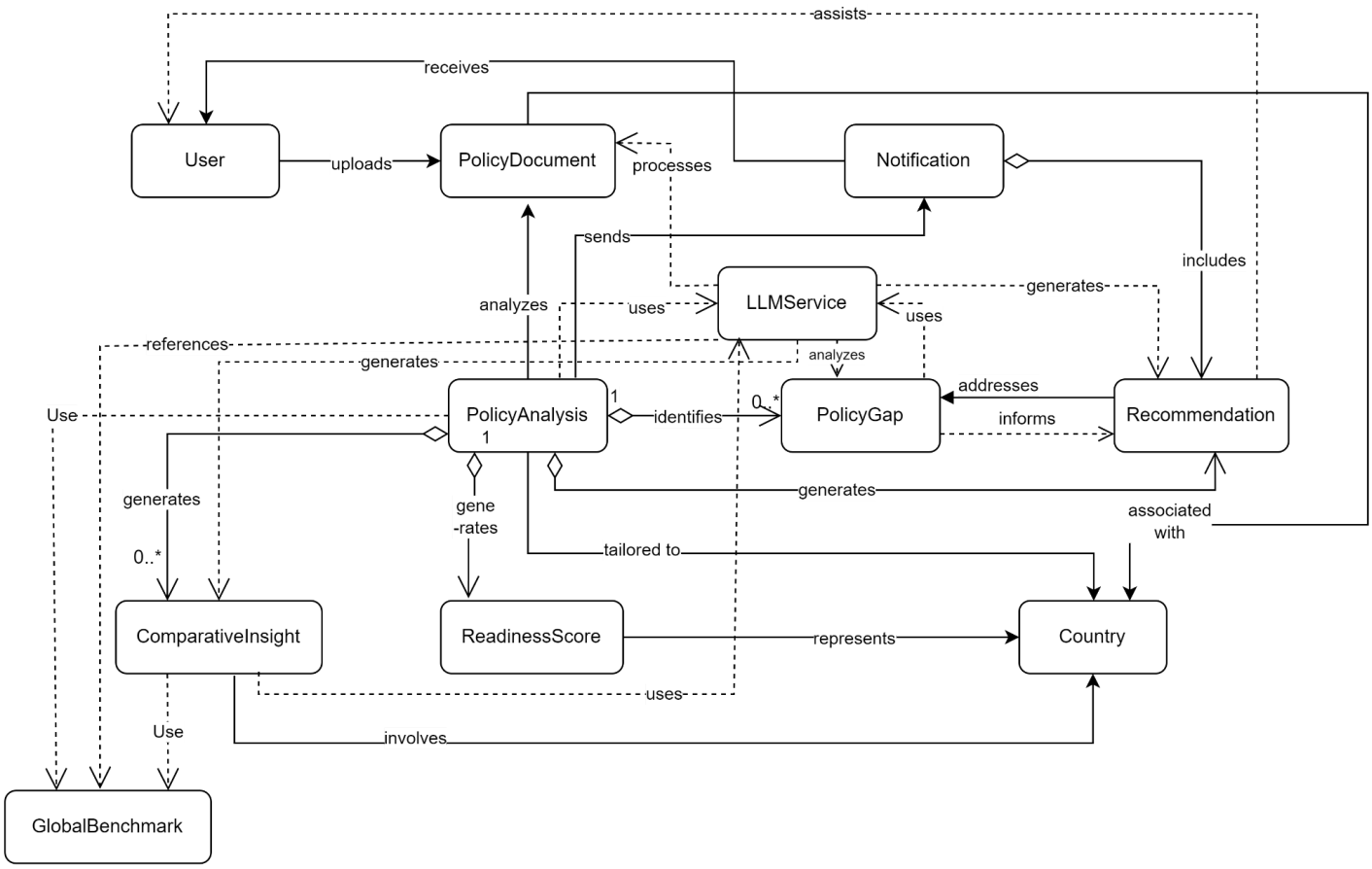


Fig-5: CRC Diagram of REAPS

6. Architectural Design

Architectural design is a visual representation in software engineering that outlines the high-level structure and organization of a software system. It focuses on defining the major components or modules of the system, their interactions, and the overall system's architecture.

6.1 Architectural Context Diagram

The architectural context diagram for the tool is given below-

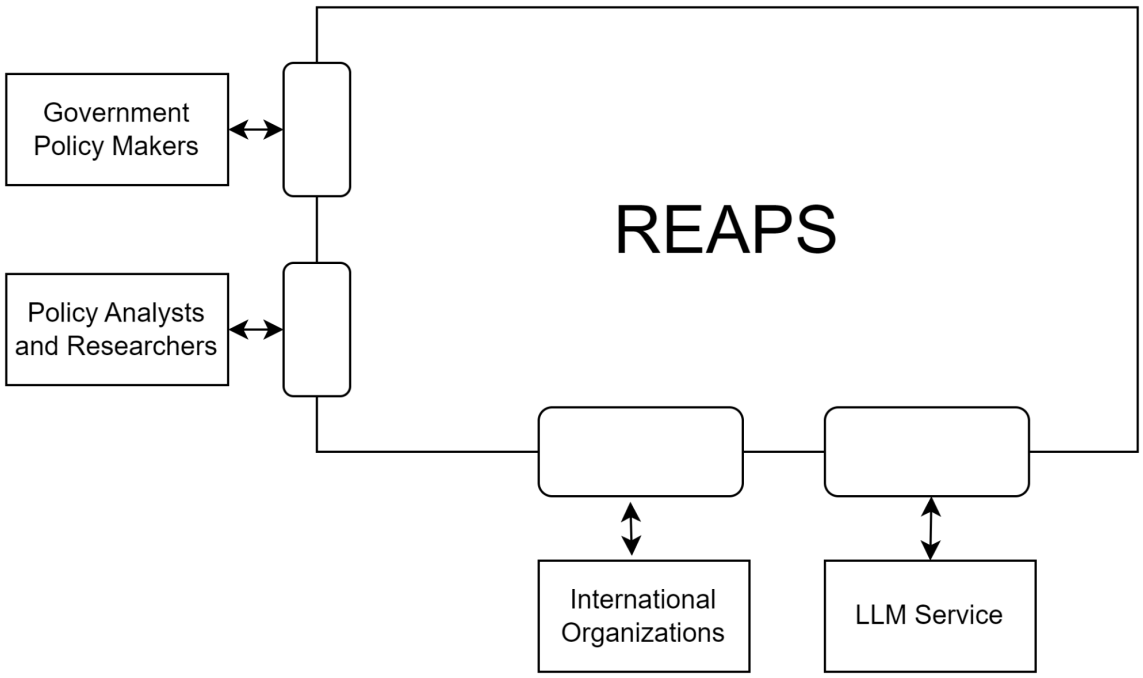


Fig-6: Architectural Context Diagram of REAPS

In the given diagram -

Users - Government Policy Makers and Policy Analysts & Researchers

Subordinate Systems - LLM Service and International Organizations

The Architectural Context Diagram for REAPS highlights its role as a decision-support tool for AI policymaking by interacting with two main user groups and integrating two subordinate systems. Government Policy Makers use REAPS to develop and refine AI policies tailored to their country's unique socio-economic and cultural contexts. Policy Analysts and Researchers leverage the tool to identify policy gaps, assess AI readiness, and align recommendations with global standards.

REAPS incorporates International Organizations (e.g., UNESCO, OECD, World Bank) as critical knowledge bases, providing AI governance benchmarks, socio-economic data, readiness indices, and global best practices to ensure its outputs are internationally relevant. Additionally, the LLM Service (Large Language Model) enhances REAPS' capabilities by processing unstructured policy documents using advanced natural language processing (NLP) techniques, extracting key insights, performing semantic analysis, and generating tailored, context-aware recommendations.

6.2 Archetypes

The archetypes for the tool are defined below-

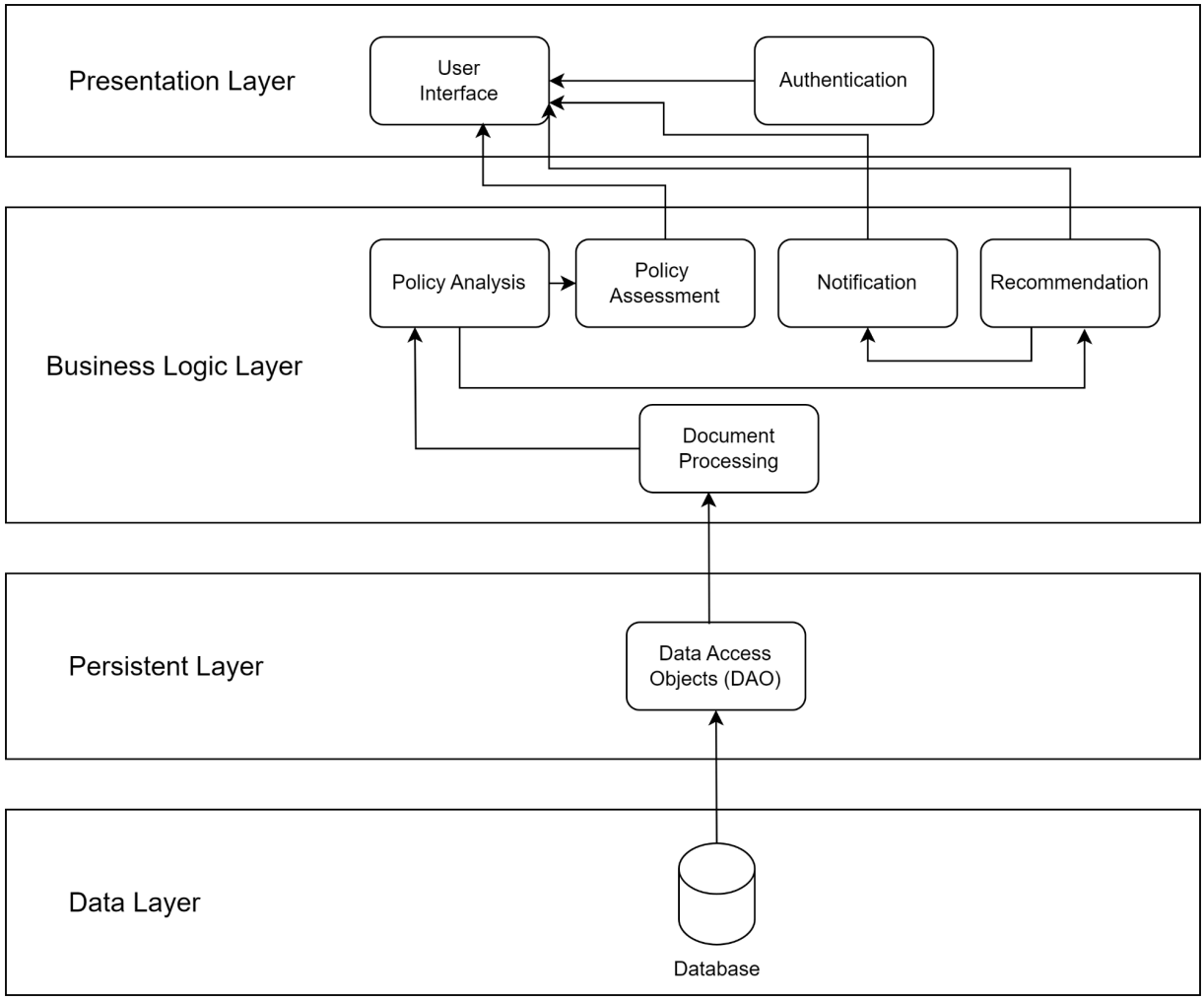


Fig-7: Archetype of REAPS

1. Data Layer

- **Database:** Central repository for storing all persistent data, including user information, policy documents, analysis results, recommendations, and feedback.
- **Data Access Objects (DAOs):** Interface layer between the application and the database.

2. Persistent Layer

- **LLM Integration Module:** Manages interactions with Large Language Models (LLMs) for natural language processing tasks.
- **Document Processing:** Handles the upload, validation, and preprocessing of policy documents using LLM.

3. Business Logic Layer

- **Policy Analysis:** Performs comprehensive analysis of uploaded AI policy documents using NLP techniques powered by LLMs.
- **Recommendation:** Generates tailored policy recommendations based on analysis results and country-specific contexts.
- **Policy Assessment:** Compares a country's AI policies with those of similar nations to provide comparative insights and validate the policy analysis
- **Notification:** Manages sending notifications and updates to users about analysis completions, recommendations, and policy trends.

4. Presentation Layer

- **User Interface:** Manages all user interactions, including uploading documents, setting preferences, and viewing results.
- **Authentication**: Handles user registration, login, and access control.

6.3 Top Level Components

Overall architectural structure with top level components is illustrated below-

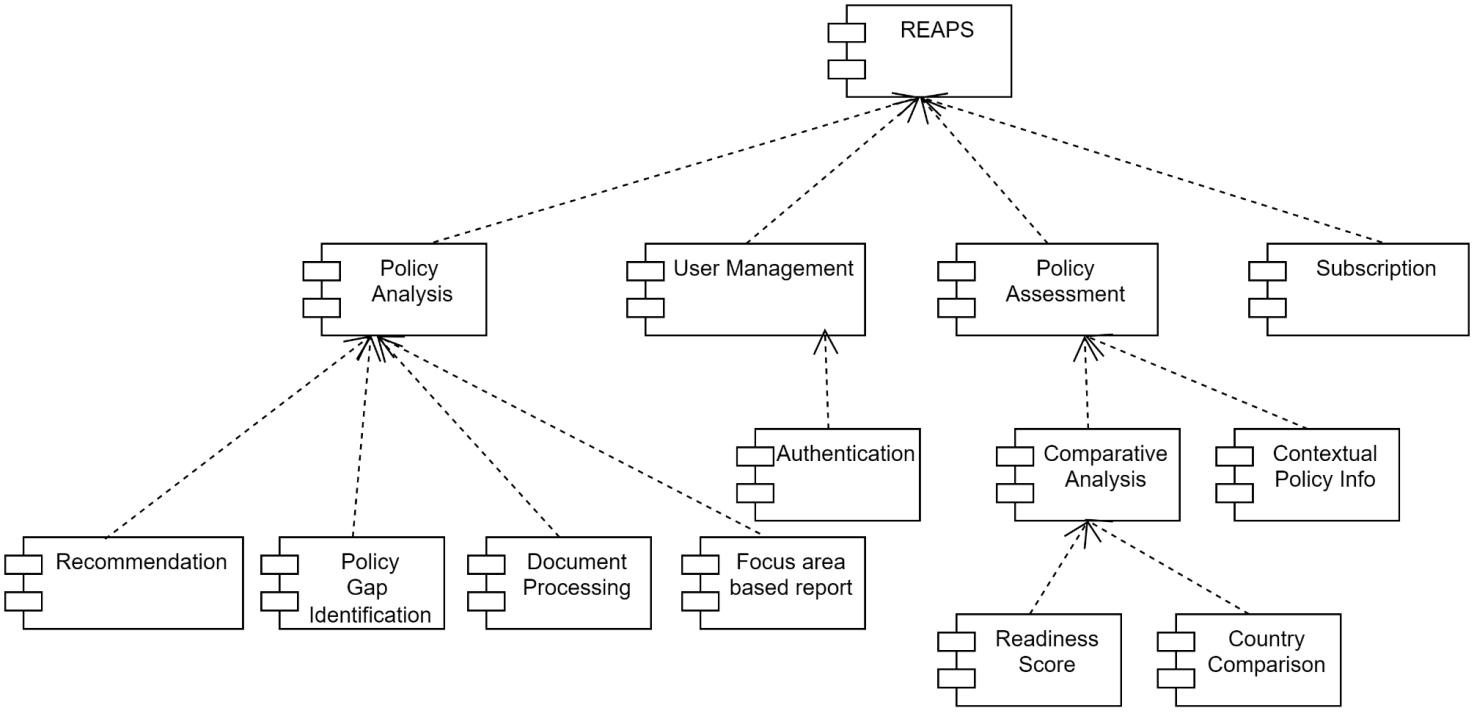


Fig-8: Top Level Components

6.4 Deployment Diagram

To effectively deploy REAPS, it's essential to have a clear and straightforward deployment architecture. This ensures scalability, maintainability, and ease of implementation. Below is a simplified deployment diagram partitioned into Frontend and Backend sections, along with External Services.

The deployment diagram of the REAPS is shown below:

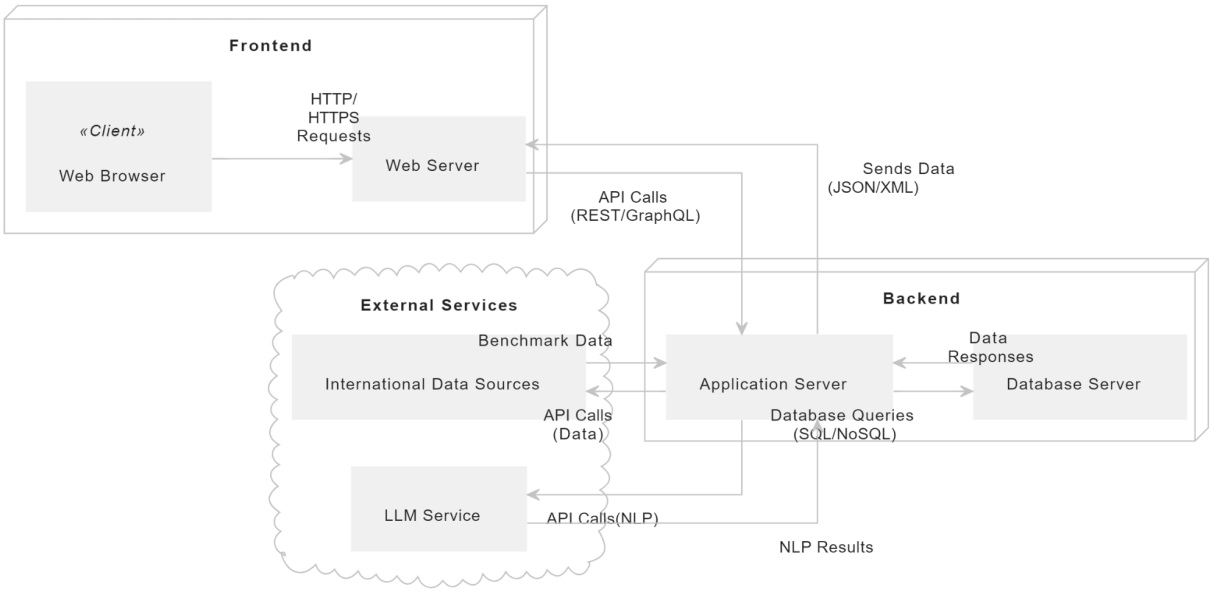


Fig-9: Deployment diagram of REAPS

7. Preliminary Test Plan

In this chapter, a high level description of testing goals and summary of features to be tested are presented.

7.1 High Level Description of Testing Goals

The primary testing goals for the REAPS system are as follows:

- To validate that REAPS meets its functional and non-functional requirements.
- To ensure the tool provides accurate and tailored AI policy recommendations based on uploaded documents and user-selected parameters.
- To verify that the NLP and ML models process, analyze, and provide results accurately without errors.
- To ensure user customization options work seamlessly.
- To test the performance and scalability of the tool under different workloads.
- To confirm the usability and accessibility of the interface across platforms.
- To validate that the feedback mechanism is functional and effective for system improvement.

7.2 Test Cases

Below is a set of test cases designed to evaluate REAPS comprehensively.

Test Case 1: User Registration and Authentication

Test Scenario: A new user registers an account and logs into the REAPS system.

Steps:

1. Register a new account and verify via email.
2. Log in using the registered credentials.

Expected Outcome:

- User account is created and activated successfully.
- User logs in and accesses the personalized dashboard.

Test Case 2.1: Document Upload and Validation

Test Scenario: A user uploads a valid AI policy document for analysis.

Steps:

1. Upload a valid AI policy document (e.g., PDF format).
2. Confirm the upload for analysis.

Expected Outcome:

- Document is uploaded and validated successfully.
- Document appears in the user's list for analysis.

Test Case 2.2: Invalid Document Upload Handling

Test Scenario: A user attempts to upload an unsupported file format.

Steps:

1. Attempt to upload a document in an unsupported format.

Expected Outcome:

- System rejects the upload.
- Error message indicates the file format is unsupported.

Test Case 3: Country-Specific Policy Analysis

Test Scenario: A user runs an analysis on an uploaded document, specifying a country.

Steps:

1. Select an uploaded document and specify the country.
2. Initiate the analysis.

Expected Outcome:

- Analysis is tailored to the selected country's context.
- Results display policy gaps and recommendations specific to that country.

Test Case 4: Receiving Tailored Recommendations

Test Scenario: After analysis, the user views the tailored recommendations provided.

Steps:

1. Access the recommendations generated from the analysis.

Expected Outcome:

- Recommendations are specific and actionable.
- Each includes details on policy areas and suggested actions.

Test Case 5: AI Readiness Scorecard

Test Scenario: The system generates an AI readiness scorecard after analysis.

Steps:

1. View the AI Readiness Scorecard associated with the analysis.

Expected Outcome:

- Scorecard displays a numerical readiness score.
- Highlights strengths and weaknesses in the current policy.

Test Case 6: Comparative Insights with Similar Countries

Test Scenario: The system provides comparative insights using socio-economic similarity.

Steps:

1. Access comparative insights after completing an analysis.

Expected Outcome:

- Displays comparisons with policies from similar countries.
- Provides validation scores indicating recommendation relevance.

Test Case 7: User Customization of Preferences

Test Scenario: The user customizes their preferences for updates and focus areas.

Steps:

1. Update preferences for peer countries, and focus areas.

Expected Outcome:

- Preferences are saved and reflected in future analyses.
- User receives confirmation of updated settings.

Test Case 8: Performance Under High Load

Test Scenario: Testing system performance when multiple users run analyses simultaneously.

Steps:

1. Simulate multiple users uploading documents and initiating analyses concurrently.

Expected Outcome:

- System handles concurrent processes without significant slowdown.
- All users receive analysis results successfully.

Test Case 9: Regular Policy Updates and Monitoring

Test Scenario: Verifying that users receive quarterly updates on AI policies and benchmarks.

Steps:

1. Ensure update frequency is set to "Quarterly."
2. Check for updates after the set period.

Expected Outcome:

- Users receive timely updates on evolving AI policies.
- Updates align with user-selected preferences.

Test Case 10.1: Validation with High-Income Countries for Similar Outputs

Test Scenario: Generate reports for multiple high-income countries and compare the similarity of recommendations to validate the system's consistency.

Steps:

1. For five high-income countries (e.g., USA, Germany, Japan, Australia, Canada), upload identical or similar AI policy documents and run analyses.
2. Compare the recommendations and analyses across these countries.

Expected Outcome:

- High similarity in recommendations and analyses.
- Recommendations reflect common socio-economic characteristics of high-income countries.
- Validates that the system correctly generalizes policies for similar socio-economic profiles.

Test Case 10.2: Validation with Low-Income Countries for Similar Outputs

Test Scenario: Ensure the system provides tailored recommendations reflecting the specific contexts of low-income countries.

Steps:

1. For five low-income countries (e.g., Ethiopia, Haiti, Nepal, Afghanistan, Madagascar), upload identical or similar AI policy documents and run analyses.
2. Compare the recommendations and analyses across these countries.

Expected Outcome:

- Recommendations are tailored to each country's unique socio-economic context.
- Differences in recommendations reflect distinct needs and priorities.
- Validates that the system does not overgeneralize and provides appropriate guidance.
- High similarity in recommendations and analyses.

Test Case 10.3: Cross-Validation Between High-Income and Low-Income Countries

Test Scenario: Compare the system's outputs between high-income and low-income countries to ensure appropriate differentiation.

Steps:

1. Use the analyses from Test Cases 10.1 and 10.2.
2. Compare recommendations and analyses between the high-income group and the low-income group.

Expected Outcome:

- Significant differences exist between the recommendations for high-income and low-income countries.
- System adjusts outputs based on socio-economic differences.
- Validates that the system differentiates appropriately between diverse socio-economic profiles.

Test Case 11: Divergence Validation for Different Income Levels

Test Scenario: Different income levels countries have different recommendations.

Steps:

- Upload AI policy documents from one high-income country and one low-income country.
- Compare the generated recommendations.

Expected Outcome:

- The recommendations should reflect significant differences, emphasizing areas of improvement for the low-income country while highlighting advancements for the high-income country.

Test Case 12: Validation Against Known Benchmarks

Test Scenario: Check if it validates the known benchmarks (e.g. OECD, UN etc)

Steps:

- Use generated recommendations for a high-income country and compare them with existing policy benchmarks (e.g., OECD standards).

Expected Outcome:

- Recommendations should align closely with global AI governance benchmarks and standards.

Test Case 13: Testing Recommendations' Customization

Test Scenario: Recommendation customization works properly.

Steps:

- Generate reports for the same country with different user-selected focus areas (e.g., ethics vs. innovation).

Expected Outcome:

- The recommendations should emphasize the chosen focus areas while retaining contextual accuracy.

8. Conclusion

In this report, I have noted down the necessary technical details required for the development of **REAPS:** Recommendation Engine for AI Adoption Policy and Strategy. The scope, assumptions, QFD, and usage scenarios are thoroughly discussed. Scenario-based modeling, data modeling, and class-based modeling have been done to define the facility and structure of the system. Besides elaborating on the architectural design, it also demonstrates the relationships between the components of the system and external systems.

It has provided a high-level background description about test goals and also about a summary of features and items, which would be tested. Images and tables have been provided throughout this document, which makes understanding a little clearer regarding these technical aspects of REAPS. While REAPS is a full-bodied tool for stringent analysis and improvement of AI policies with respect to global standards, there is still some scope for betterment. The future scope could bring the inclusion of more subordinate systems, added customization areas for policy focus, and further advanced AI models to enable more specific and adaptive suggestions.

The document lays a formal overview of the flow of REAPS with it, so that the reader has a complete understanding of the technical design and working of the system. It is expected that this report will be very useful for stakeholders and developers working in the tool.

References

1. https://www.oecd.org/en/topics/policy-issues/artificial-intelligence.html
2. Igbinenikaro, Etinosa and Adefolake Olachi Adewusi. “Policy recommendations for integrating artificial intelligence into global trade agreements.” International Journal of Engineering Research Updates (2024): n. pag.
3. https://artificialintelligenceact.eu/assessment/eu-ai-act-compliance-checker/